EXP-8

06/03/26

[Transient Analysis of Series RLC circuit]

Aim :- To determine transient response of series RC circuit using LT Spice Simulator.

Apparatus required :-

S.NO	Components	Specification	Quantity
1.	CRO	10 MHz	1
2.	DCB	–	1
3.	Resistor	10k Ω	1
4.	Breadboard	–	–
5.	Connecting Wires	–	–
6.	LT Spice Simulator	–	–

Theory :-

When an increasing DC voltage is applied to a discharged capacitor, the capacitor draws a charging in the current and "charges up" and when the voltage is reduced, the capacitor's energy is the opposite direction, because capacitors are such use.

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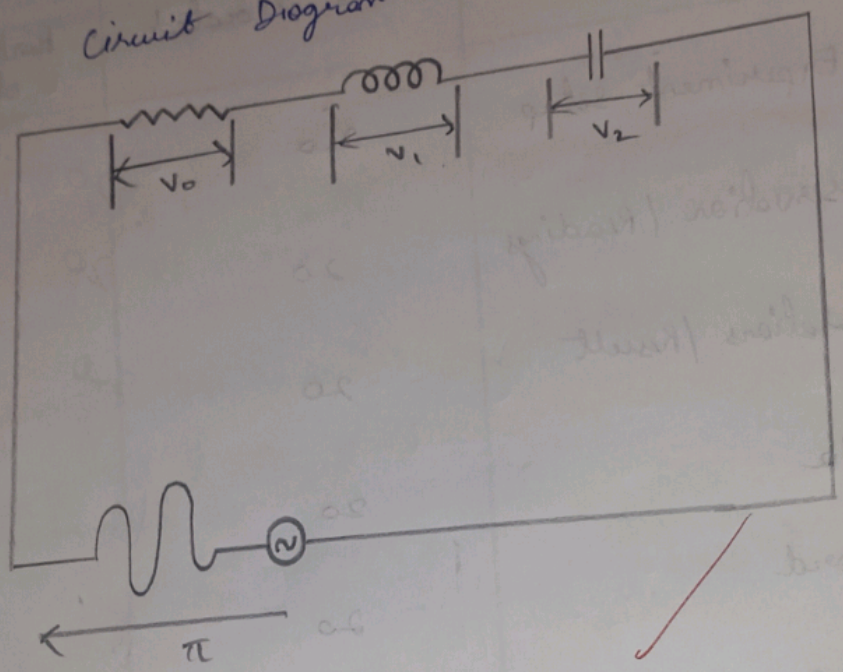
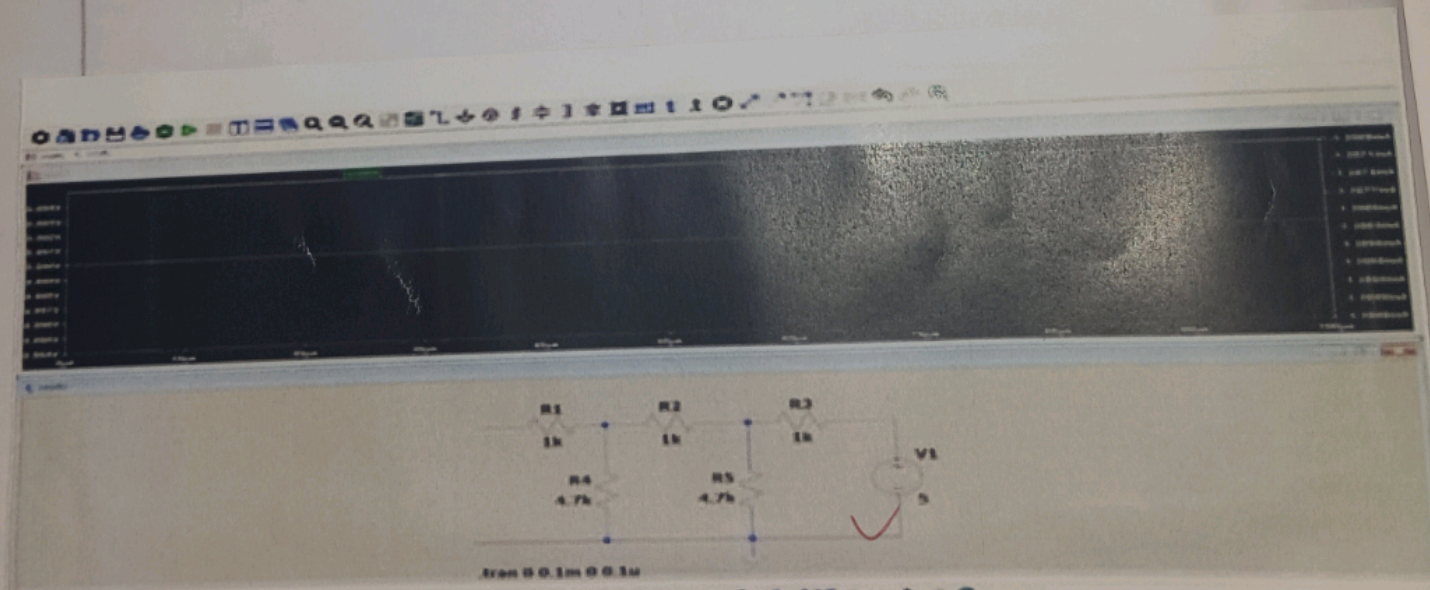
Circuit Diagram :-





Tabular column :

Time	$V_L(t)$ - voltage across inductor observed value
$t=0$	1.16 V
$t=\pi$	6.34 V
$t=2\pi$	8.66 V
$t=3\pi$	9.44 V
$t=4\pi$	9.18 V
$t=5\pi$	9.55 V

Procedure :-

1.) Make connections as per the circuit diagram.

2.) calculate the values of $V_c(t)$ and write table and compare the calculated values and measured values of voltage in both the cases.

S.NO	Description	Marks Alloted	Marks obtained
1.)	Experiment	20	20
2.)	observation	20	18
3.)	calculation	20	18
4.)	Viva	20	18
5.)	Record	20	20
6.)	Total	100	94

Result :-

Thus the transient analysis of series RCL circuit is done practically and verified with theoretical value using LTSpice Simulator.